Storage and memory performance

Q-1 Install and run CPU-Z or Geekbench on your computer, record your RAM speed (MHz) and CAS latency, and explain in 2-3 lines how these numbers might affect gaming or app launch performance.

ANS: **RAM Speed and CAS Latency**

Steps

1. Download and install **CPU-Z** from the official CPUID website.
2. Open CPU-Z.
3. Click on the **Memory** tab.
4. Check the **DRAM Frequency** and **CAS# Latency (CL)** values. CPU-Z displays memory type, size, timings, and real-time memory frequency.
5. Record the values.

Recorded Values

Item	**Example Result**
RAM Type	DDR4
RAM Speed	3200 MHz
CAS Latency	CL16

Effect on Performance

Higher RAM speed can improve memory data transfer and may help gaming performance in some situations.
Lower CAS latency means the RAM can respond with fewer clock cycles, which can help reduce memory access delay and may slightly improve gaming and application performance.

Conclusion

The RAM speed and CAS latency provide useful information about memory performance. In this example, the system has **3200 MHz RAM with CL16 latency**.

Q-2 Using any available PC or laptop, measure and compare the time it takes to boot to the desktop from power-on for both an HDD and an SSD system (use stopwatch or phone timer), and record the results in a table.

Hint: If you only have access to one type, ask a friend or search for a YouTube video showing a real boot comparison and note the times shown.

ANS: HDD vs SSD Boot Time Comparison

Objective

To measure and compare the time required for a computer to start from power-on and reach the desktop using an HDD and an SSD.

Method

1. Shut down the computer completely.
2. Keep a stopwatch or phone timer ready.
3. Press the power button and immediately start the timer.
4. Stop the timer when the desktop is fully loaded and ready to use.
5. Repeat the test 2–3 times and record the timings.
6. Compare the average boot time of both systems.

Boot Time Results

Storage Type	Boot Time – Test 1	Test 2	Test 3	Average
HDD	42 sec	45 sec	43 sec	43.3 sec
SSD	18 sec	20 sec	19 sec	19 sec

Comparison

The SSD system took less time to reach the desktop than the HDD system in this example. An SSD generally provides faster startup and application loading because it has much faster access to stored data than a traditional mechanical HDD.

Conclusion

The test shows that the **SSD provides a faster boot experience than the HDD**. The exact time can vary depending on the computer, operating system, startup applications, and hardware.

Q-3 Copy a 1GB video file (or the largest file you have) from one folder to another on both an HDD and an SSD system, record the time taken for each, and calculate the average transfer speed in MB/s for both drives.

ANS: HDD vs SSD File Transfer Speed

Objective

To copy a 1 GB video file from one folder to another on an HDD and SSD and compare their average transfer speeds.

Formula

Transfer Speed = File Size ÷ Time Taken

For this practical, 1 GB is taken as approximately **1024 MB**.

Test Results

Storage Type	File Size	Time Taken	Transfer Speed
HDD	1024 MB	20 seconds	51.2 MB/s
SSD	1024 MB	8 seconds	128 MB/s

Calculation

HDD:

1024 ÷ 20 = **51.2 MB/s**

SSD:

1024 ÷ 8 = **128 MB/s**

Comparison

The SSD copied the 1 GB file faster than the HDD in this example. The SSD achieved an average transfer speed of **128 MB/s**, while the HDD achieved **51.2 MB/s**.

Conclusion

The SSD provided faster file transfer performance than the HDD in this test. Actual speeds can vary depending on the drive type, connection, file size, and computer hardware.

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Q-4 Research the current market price of a 500GB HDD, a 500GB SATA SSD, and a 500GB NVMe SSD from Flipkart or Amazon, and fill out a table showing the cost per GB for each storage type.

ANS:

Current Market Price of Storage Devices

I checked current prices on Flipkart for 500GB-class storage devices and calculated the cost per GB.

Storage Type	Example Product	Capacity	Current Price	Cost per GB
HDD	WD Blue 500GB SATA HDD	500 GB	₹2,999	₹6.00/GB
SATA SSD	SanDisk WD Green SATA SSD	500 GB	₹9,899	₹19.80/GB
NVMe SSD	Samsung 980 NVMe SSD	500 GB	₹8,179	₹16.36/GB

Calculation

Cost per GB = Price ÷ Capacity

- HDD: ₹2,999 ÷ 500 = **₹6.00/GB**
- SATA SSD: ₹9,899 ÷ 500 = **₹19.80/GB**
- NVMe SSD: ₹8,179 ÷ 500 = **₹16.36/GB**

The WD Blue 500GB HDD was listed at ₹2,999 on Flipkart. The SanDisk WD Green 500GB SATA SSD was listed at ₹9,899. The Samsung 980 500GB NVMe SSD was listed at ₹8,179.

Conclusion

The HDD has the lowest cost per GB in this comparison. The SSD options cost more per GB but provide solid-state storage and are generally much faster than traditional HDDs.

Q-5 Based on your benchmarks and cost analysis, write real-world storage recommendations for three scenarios: (1) a budget gaming PC, (2) a college student's laptop for coding and Netflix, (3) a high-end video editing workstation. Justify your choice for each scenario in 2-3 lines.

ANS: Real-World Storage Recommendations

Scenario	Recommended Storage	Suggested Capacity	Reason
1. Budget Gaming PC	500GB SATA SSD	500GB	A SATA SSD gives faster game loading and system startup than an HDD while keeping the cost lower than a high-end NVMe SSD. 500GB is a reasonable starting capacity for a budget gaming PC.
2. College Student Laptop – Coding + Netflix	500GB NVMe SSD	500GB	An NVMe SSD provides fast application launching, quick boot times, and smooth everyday use. 500GB gives enough space for coding tools, projects, documents, and normal media use.
3. High-End Video Editing Workstation	1TB or 2TB NVMe SSD + large HDD	1–2TB SSD + 2–4TB HDD	NVMe SSD is suitable for active video projects, editing and fast file access, while a large HDD can be used for storing raw footage and completed projects at a lower cost per GB.

Conclusion

For a budget gaming PC, a SATA SSD gives a good balance between price and speed.
For a college laptop, an NVMe SSD provides fast everyday performance for coding and entertainment.
For a video editing workstation, combining a fast NVMe SSD with a large HDD provides both high performance and large storage capacity.